

APPENDIX

In this extension of our main report, we provide additional experiment and metric plots using Gaia that do not fit into the main page limit.

A. Network Round Trip Times

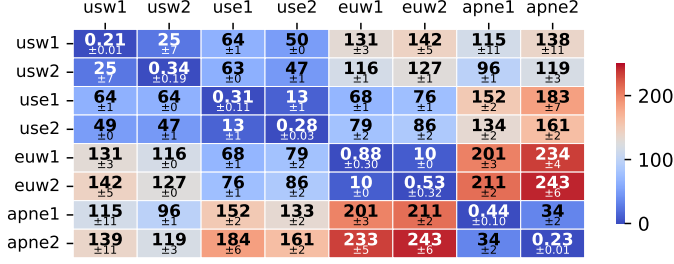


Fig. 11. Average round-trip times (ms) and standard deviation (ms) for each region pair used across 3 runs.

Figure 11 depicts the latency and standard deviation under normal experiment conditions.

B. Latency Ablations

Figure 12 showcases how the latency evolves for different types of transactions as we make the TPC-C workload more geo-distributed.

C. Resource Utilization

In Figure 13, we once again use the TPC-C workload to demonstrate how many resources are needed by different systems, and how many aborts are triggered.

D. Access Skew

In Figure 14, we plot the performance under the access skew scenario.

E. Network Latency & Jitter

In Figure 15, we plot the resource utilization and abort rate under the network latency and jitter scenario.

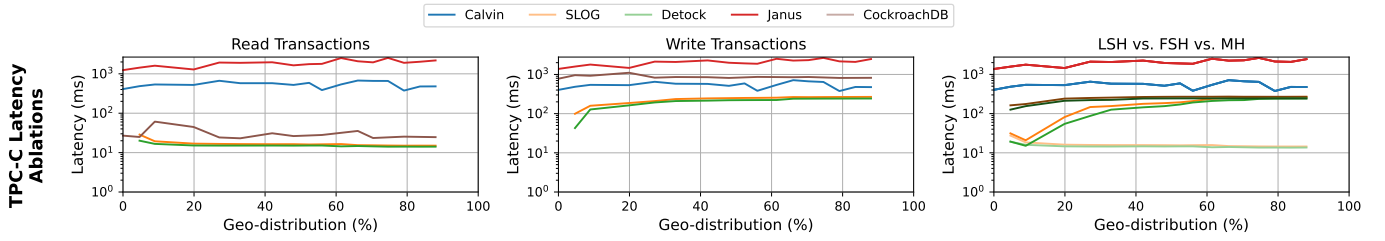


Fig. 12. The TPC-C read vs. write and LSH vs. FSH vs. MH latency ablations under the access patterns scenario. In the LSH vs. FSH vs. MH subplot the lightest shades represent LSH, medium shades represent FSH and darkest shades represent MH transactions.

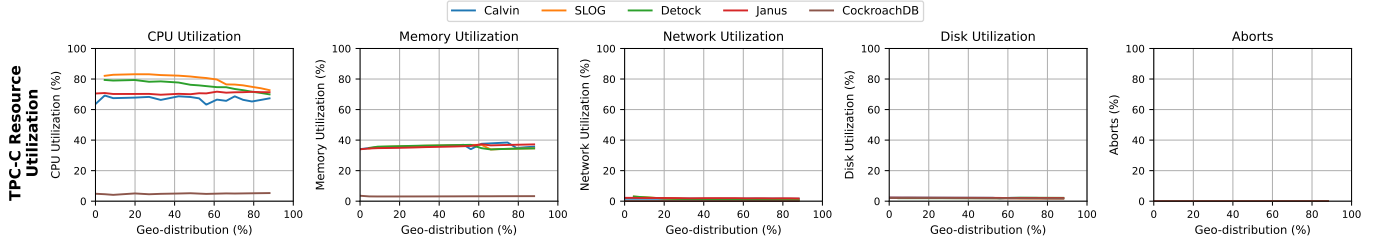


Fig. 13. The TPC-C CPU, memory, network, and disk utilization, and abort rate under the access patterns scenario.

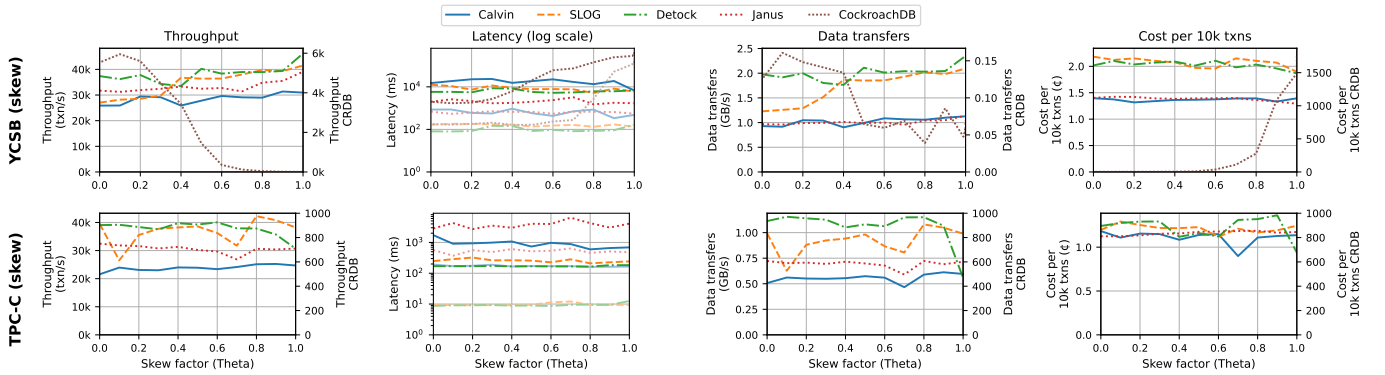


Fig. 14. The YCSB and TPC-C throughput, latency, data transfers, and cost per transaction under access skew. Opaque lines in the latency figure represent p50 and bold ones p99.

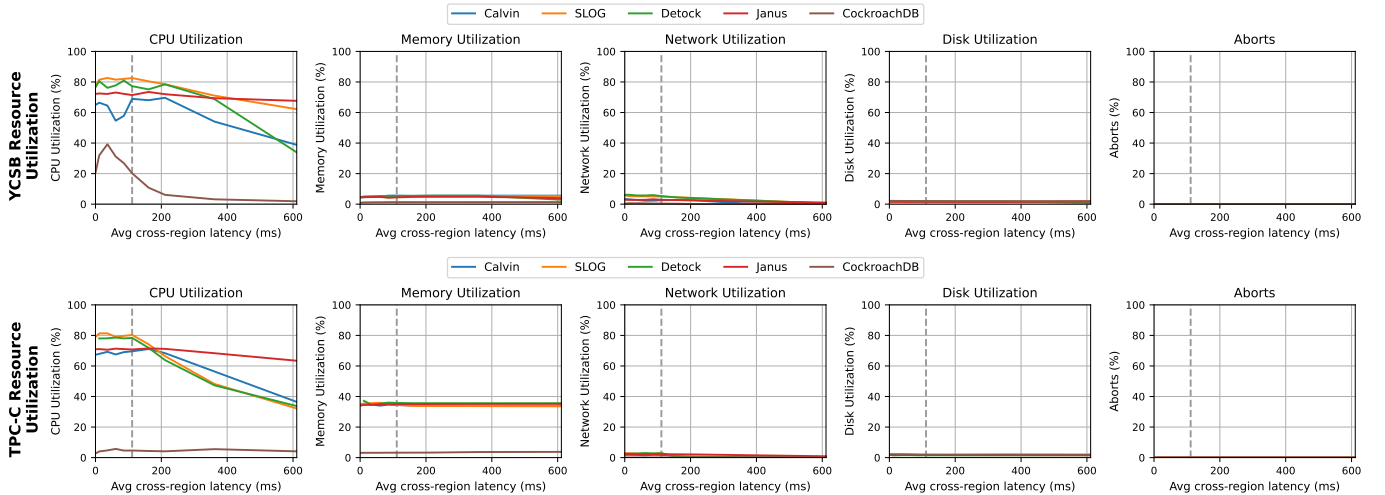


Fig. 15. The YCSB and TPC-C CPU, memory, network, and disk utilization, and abort rate under extra network latency and jitter. The dashed vertical line represents the average cross-region latency in the default geo-distribution setup.